AP ELECTRIC · AP UNO: BASE + CORE en una placa. I2C: INA238 0x40, TMP1075 0x48, M24C02 0x50, salidas TCA9554 0x20 (S1–S4, P5=0 / P6=1 identificación, P7 = DE RS-485), entradas TCA9554 0x24 (E1–E4). RS-485: TX IO1, RX IO3 (DMX512 o Modbus RTU).

PWR_FLAG VIN_RAW PWR_FLAG GND PWR_FLAG VIN PWR_FLAG GNDIN PWR_FLAG V12 PWR_FLAG 3V3 PWR_FLAG VF PWR_FLAG VCC PWR_FLAG VLOG PWR_FLAG

J1 ENTRADA 12–24V
F1 MINI 20A
RS1 1mR 2512 2W
U1 INA238
C4 100nF
QR1 BSC016N06NS
RG0 10k
DZ1 MMSZ5242B 12V
D1 SMBJ26A

C1 330uF 50V
C3 10uF 50V
C13 10uF 50V
J3 +V SALIDA
TP1 VIN
U2 LMR16006XDDC
L1 47uH
D7 SS210
C5 100nF

C6 10uF 50V
C7 22uF 25V
RF1 150k
RF2 10k
TP2 12V
F2 3A
J5 FOCO CC 1
J8 FOCO CC 2
U6 AL8860MP

RSC1 0R1
L2 47uH
D2 SS34
C11 4.7uF 50V
RC1 2.2k
U7 AL8860MP
RSC2 0R1
L3 47uH
D3 SS34

C12 4.7uF 50V
RC2 2.2k
U3 UCC27524D
U4 UCC27524D
D8 1N4148WS
D9 1N4148WS
C8 1uF 25V
C9 1uF 25V
J4 CANALES

Q1 BSC028N06LS3
RG1 4.7
RP1 10k
Q2 BSC028N06LS3
RG2 4.7
RP2 10k
Q3 BSC028N06LS3
RG3 4.7
RP3 10k

Q4 BSC028N06LS3
RG4 4.7
RP4 10k
U8 TMP1075D
C15 100nF
U9 M24C02-WMN6TP
C14 100nF
TP3 3V3
TP4 GND

U10 ESP32-C3-WROOM-02F
C20 12pF
C21 100nF
R20 10k
C22 1uF
R21 10k
SW1 MODO / BOOT
SW2 RESET
R22 4.7k

R23 4.7k
R24 10k
R25 1k
D12 ROJO
R26 1k
D13 VERDE
R27 10k
R28 10k
R29 10k

J10 USB-C
R30 10k
R31 5.1k
R32 5.1k
U11 USBLC6-2SC6
D10 B5B19W
D11 B5B19W
U12 AP63203WU
L4 4.7uH

U5 TCA9554PWR
U13 TCA9554PWR
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4 x 8 A + 2 focos CC + 4 salidas + 4 entradas + RS-485 + Wi-Fi
Placa de 4 capas, 2 oz exteriores (JLCPCB)
PCB 120 x 56 mm, 4 capas

Sheet: /
File: AP_UNO.kicad_sch

Title: AP ELECTRIC · AP UNO — módulo compacto todo en uno

Size: A2	Date: 2026-10-11	**Rev: 1.0**
KiCad E.D.A. 9.0.9		Id: 1/1